THE KOSELLECK MACHINE

Testing the Sattelzeit as a Network Event

A method note on the Koselleck Machine: measuring whether English vocabulary reorganized as a system, and not only word by word, across 1770-1830.

REPOSITORY	koselleck-networks
DOCUMENT	docs/method.tex → docs/method.pdf
CORPUS SPAN	1510-1910, uniform 20-year windows
DISPLAY RESOLUTION	per-variant, auto-picked (1,500-word cap); sweep: 15 settings
COMPILED	September 9, 2026

Contents

1 Context

The historian Reinhart Koselleck argued that the period 1770-1830 (the **Sattelzeit**, German for "saddle period") was not simply an era in which many words happened to change meaning at the same time. He claimed it was a moment when the whole vocabulary of politics and society reorganized *as a system*: concepts shifted together, in relation to one another, rather than drifting independently one at a time. This has always been argued in prose, from close reading of texts. It has never been tested computationally.

Ryan Heuser's *Computing Koselleck* took the first step towards such a test: he trained a separate word-embedding model for each historical period and measured how far each individual word moves from one period to the next. He found that this movement is unusually large around 1770-1830, which supports the idea that something real happened in that window. But his method looks at one word at a time. It cannot say whether words moved *together*, as a system reorganizing around a shared structure, or whether each word's shift is its own independent event that simply happens to cluster in time.

This project adds that second, missing test. Instead of asking how far a single word moves, it asks whether the groupings among words, the way the whole vocabulary is organized into clusters of related meaning, change unusually around the same decades. If Koselleck is right, we should see the cluster structure itself reorganize at the Sattelzeit, not just individual words drifting off on their own.

One real example from the built data shows what this looks like concretely. In the network for 1770-1790, *nation* sits in a community of civic-administration words (*treasurer, sheriffs, electors, burgesses*), while *revolution* sits in an unrelated community of shape and measurement terms (*oval, oblong, compressed, bushels*). By the very next window, 1790-1810, both words - along with *constitution* - have moved into a single new community built around abstract systemic and governmental process words (*government, arrangements, processes, phenomena*). Three politically central words, previously scattered across unrelated communities, land in the same one at exactly the transition into the French Revolutionary period. This is one concrete instance of the pattern the **migration fraction** (item 6 below) measures in aggregate across the whole vocabulary - not proof on its own, but a picture of what "moved together" actually looks like in the data.

2 Method

The method reuses the same per-period training that Heuser's approach uses, so the two studies remain directly comparable, and adds a network layer on top that a word-by-word approach cannot produce.

1. **Split the corpus into time windows.** The corpus is divided into even 20-year windows, currently 1510 to 1910, starting at 1510 rather than 1500 so that both 1770 and 1830 land exactly on a period boundary - the transitions that open and close the Sattelzeit are each a literal edge between two periods, not an approximation. The windows are uniform across the whole timeline, not just around the Sattelzeit: measuring more finely right where reorganization was expected and more coarsely everywhere else would bias the method toward finding exactly what it was looking for. Heuser's own results show a second spike of word-level change around 1905-1925, unrelated to the Sattelzeit, a pattern a Sattelzeit-only sampling scheme could never have picked up. The span reflects the corpus assembled for this project so far; the same pipeline applies unchanged to any other span or window size, set entirely in configuration, given a period-dated corpus to feed it.

2. **Train a language model per period.** For each time window, a model (**word2vec**) is trained independently, learning a numeric representation of every word from how it is actually used in that period's text: words that appear in similar contexts end up with similar representations. For example, if "king" and "queen" both tend to appear near words like "crown" and "reign" in a period's text, the model gives them similar numeric representations. Training a separate model per period, rather than one model updated continuously, is what makes it possible to later measure both individual word drift and whole-system reorganization: a word's meaning only makes sense relative to the other words trained alongside it, in that same period.

3. **Build a word-similarity network.** From those representations, a network is built for each period: every word is a node, linked to its 15 closest neighbours by **cosine similarity** (a standard measure of how alike two words' representations are). A fixed number of nearest neighbours is used instead of a similarity threshold, because at this vocabulary scale a fixed threshold produces networks too dense to cluster meaningfully: in early testing, on a then-smaller corpus, a 0.3 threshold linked roughly one in ten possible word pairs, tens to hundreds of millions of edges per period - a density that only gets worse now that per-period vocabularies reach up to 170,000 words.

   A fixed, hand-curated list of about ninety common function words - articles, prepositions, conjunctions, the standard pronoun forms, auxiliary verbs, plus early-modern spelling variants (*hath*, *doth*, *thou*, *ye*, *unto*) that EEBO/ECCO/Evans spelling keeps unmodernized - is excluded from the network as nodes, along with every single-letter token (always either one of these words, a Latin citation abbreviation, or an OCR artifact).

These words still take part in training, where they help place every other word's meaning correctly; they are left out only at this network-building stage, since the interest here is in words that carry conceptual content. The exclusion is this specific list, not a part-of-speech rule - it does not remove every pronoun as a grammatical category, only the particular forms named above.

4. **Detect communities.** A **community detection** algorithm (**Leiden**) is run on each network, grouping words that are densely connected to one another. In practice, these groups function as semantic fields: sets of words that, in that period, are used in similar ways to each other more than to the rest of the vocabulary.

   This differs from other clustering methods in a basic way. Some methods, k-means among them, fix a number of groups in advance and assign each item to whichever group's average position it sits closest to. Leiden does neither: it does not fix the number of groups beforehand, and it never compares a word to a group's "average", it only ever asks a relational question about a word's actual connections. For any word, and any group it is already directly connected to, Leiden checks whether moving that word into the group would make the network as a whole denser *within* groups and sparser *between* them than plain chance would predict, given how connected that word and that group already are overall. If the move helps, it is made; if not, the word stays put or a different move is tried. Repeating this, word by word, until no single move can improve things any further, is the entire algorithm.

   The number this process is trying to improve at every step has a name, **modularity** (written Q). It compares how connected the words placed inside the same group actually are against how connected they would be expected to be if the same set of connections had been handed out at random instead. A grouping that captures real structure scores well above zero; a grouping no better than chance scores around zero; a grouping that forces genuinely unrelated words together scores *below* zero, actively worse than not grouping at all. Leiden's search is nothing more than hill-climbing on this one number.

5. **Repeat detection at several levels of detail.** Community detection does not have one single "correct" answer: a parameter called **resolution** controls how many groups appear and how large they are. Raising resolution produces more, smaller, more tightly-defined groups; lowering it produces fewer, larger, more loosely-defined ones - by changing how large a word's own connections have to

be, relative to chance, before joining a group is judged worthwhile. For that reason every network is processed at fifteen fixed resolution settings, currently 0.1 to 16.0, and any claim about reorganization has to hold up across all of them, not just one chosen after the fact.

Hard rule. No single setting among the fifteen is ever treated as "the" result and the rest discarded. The sweep's entire purpose is to check that the same conclusion holds broadly across settings, not to pick whichever one happens to agree with the hypothesis. No finding is reported if it only holds at a single, convenient resolution.

These fifteen values are calibrated to this project's current corpus, not a fixed universal range: the sweep has already been extended once before as the corpus grew, raised in two steps to reach today's ceiling of 16.0 once the British Library supplement pushed per-period vocabularies past what the earlier range could meaningfully resolve (that extension's full detail lives in "Choosing the labeling resolution" below, alongside the related display-resolution history). A substantially larger future corpus would likely need this range extended again the same way.

A second, separate resolution decides only which single partition is displayed and labeled in the interactive explorer below - a reading convenience, with no bearing on which finding gets reported. It is picked per period and per region automatically, by a bisection search independent of the sweep's fifteen values - starting from resolution 1.0 and doubling or halving to bracket, then narrowing, the lowest resolution at which no community grows past a 1,500-word cap. Unlike the sweep, it has no fixed ceiling or floor: a given period and region can land above 16.0 or below 0.1. "Choosing the labeling resolution" below covers the mechanism and why it replaced an earlier fixed choice.

6. **Compare consecutive periods.** For each pair of consecutive periods, the comparison is restricted to words that appear in both (each period has its own vocabulary). The best possible match between one period's groups and the next period's groups is found first, because group numbers do not mean the same thing from one period to the next, so the correspondence has to be established before anything can be compared.

   Concretely (`metrics.py`'s `align_communities`): build a contingency table (a grid of counts crossing every old group against every new group) counting, for every (old group, new group) pair, how many shared words fall in that pair, then solve it as a single global assignment problem (the **Hungarian algorithm**, a standard way to find the best one-to-one matching between two groupings, run via SciPy) that picks the one-to-one relabeling of new group numbers onto old ones maximizing the total word count matched. This is a fixed, deterministic optimum over all groups at once, computed once per period pair – there is no percentage-overlap threshold anywhere in it, and no per-group decision made independently of the others.

A single shared word then counts as *moved* if, under that one relabeling, its new group's mapped-back old identity differs from the old group it was actually in; otherwise it counts as *stayed*. The fraction of shared words that still ended up in a different group, despite that best match, is what this project calls the **migration fraction**. For example, if 100 words appear in both period A and period B, and 40 of them end up in a different group even after the best possible matching, the migration fraction for that pair is 0.40. It is this project's central measurement of how much reorganization happened between two periods, computed once per period pair at each of the fifteen sweep resolutions – this is the figure item 7 tests the hypothesis against.

The interactive explorer (Section 3) shows a second, separate version of the same measurement, computed the identical way but between each period pair's own auto-picked **display resolution** (item 5) rather than a shared sweep value – since different periods can land on different display resolutions, this is its own alignment, not a lookup into the sweep result. This is the number, and the per-word "moved"/"stayed" arrow, that belongs next to a community a reader is actually looking at; it is never what a reorganization claim is tested against, which is always the sweep-based figure above.

7. **Test whether reorganization concentrates at the Sattelzeit.** The hypothesis predicts that the migration fraction should be unusually high specifically for the transitions that fall inside or close out the 1770-1830 window, compared to every other transition across the full 1510-1910 timeline.

   Result. Averaged across the fifteen-resolution sweep, the transitions bracketing the Sattelzeit do rise – combined region: 1750-1770 to 1770-1790 at 0.41, 1770-1790 to 1790-1810 at 0.45, 1790-1810 to 1810-1830 at 0.52 – but this is a gradual climb, not a spike isolated to the window: 1810-1830 to 1830-1850 stays elevated at 0.50 rather than falling back afterward, and the single highest transition in the entire 1510-1910 series falls before the Sattelzeit, not inside it. For a sense of scale, the migration fraction across the full 1510-1910 series spans a range from transitions where the partition stays roughly consistent to ones, like the pre-Sattelzeit peak below, where more than half of the shared vocabulary changes group; the Sattelzeit-adjacent numbers above sit toward the high end of that range without being its maximum. (Exact figures below are the fifteen-resolution average for one pipeline run; see Reproducibility under Current limitations for how much they can be expected to shift on an independent rerun.)

   1690-1710 to 1710-1730 measures 0.55 in the combined corpus and 0.62 in the British-only split (see the region-split check below) – both above every transition the Sattelzeit window itself produces. That pre-Sattelzeit peak coincides with a real, isolated drop in shared vocabulary between those two periods, consistent with a corpus-coverage artifact – ECCO-TCP thins out precisely in this range (see Corpus below) – rather than a genuine reorganization signal, though this has not been confirmed either way.

Taken together, this measurement does not show a clean, isolated reorganization spike at 1770-1830: the pattern is a modest, gradual rise that survives the region-split check but does not stand out from at least one earlier, likely-artifactual transition. Full per-resolution numbers and the complete period series are in `docs/pipeline_manual.pdf`'s Stage 8 section.

8. **Repeat the whole measurement per region.** The corpus is not evenly British across the timeline, and the imbalance is worst exactly where it matters most (the real numbers are in "Corpus coverage" under Current limitations, below): a change in the cluster structure at that boundary could therefore reflect a change in *who was writing* rather than a change in the language.

   To separate the two, every step above (training, network construction, community detection, the resolution sweep, and the migration fraction) is also run on each region's documents alone, in addition to the combined corpus. If the reorganization is real, it should show up inside the British-only and the American-only timelines as well, not only in the mixture of them; if it shows up only in the combined run, it is an artifact of corpus composition rather than semantic history. Nothing here hardcodes which regions exist: a document's region is simply the top-level folder its source was filed under, so the same check extends to any other split of the corpus.

2.1 Corpus

Four sources were considered for the corpus; only the first two are actually used:

- **Primary source.** The **Text Creation Partnership (TCP)**: curated, manually corrected transcriptions with no OCR noise, covering 1500-1800 (EEBO-TCP, ECCO-TCP, Evans-TCP).
- **Supplement.** Since the Sattelzeit extends to 1830 and TCP stops at 1800, the 1800-1900 gap is filled by the **British Library**'s digitised 19th-century books collection (49,455 books, roughly 65,000 volumes, OCR-derived text with catalogue metadata already attached, public domain).
- **Considered and rejected.** Project Gutenberg was considered first and dropped: its metadata carries only its own digitisation date, not the original print publication

date, which makes it unusable for a method that buckets every document by the year it was actually written.

- **Considered, not used.** HistWords (pre-trained historical embeddings) was identified as a possible independent check against this project's own result, but no such comparison has actually been run - nothing in the pipeline reads or depends on it.

Each source is filed under a region: `british` for EEBO-TCP, ECCO-TCP, and now the British Library supplement; `american` for Evans-TCP, which stops at 1800 along with the rest of TCP. The region tag is not a claim about two different English-language communities - for most of this timeline, American print culture and British print culture share the same language, and treating them as linguistically distinct would not be defensible.

It tracks archive provenance instead: EEBO/ECCO and the British Library are one continuous British-print lineage, Evans is an independently curated American-print archive. As item 8 above explains, that is what makes the region split a robustness check rather than a dialect comparison; the combined corpus stays the headline result either way. A direct consequence of this design: since the British Library supplement is tagged `british`, the `american` region has no data past 1800 at all, an honest asymmetry rather than a gap to paper over.

2.2 Not a closed dataset

This project is a method. The corpus is one input to it, fetched separately and deliberately not vendored: a reader who disagrees with it, or has a better one, can rerun the whole measurement on their own text rather than take these numbers on trust. Swapping in a different corpus is either a folder drop (already-valid TCP P4 XML needs no code change) or one new reader function beside `iter_xml_members` in `src/parse_tcp.py`, emitting the same `{region, source, doc_id, year, text}` record the TCP path already writes - everything downstream reads only that normalised record and needs no change. README's Adding a new corpus section gives the full worked contract, including the exact JSON shape and the two concrete cases. There is no upload interface, hosted service, or API-key-driven ingestion: clone the repository, put corpus files in the right folder, rerun the pipeline scripts in order.

3 The interactive explorer

The historical argument is complete at this point: every quantitative claim above, including whether reorganization concentrates at the Sattelzeit, is produced by `src/metrics.py` and needs no labels or web interface at all. What follows describes optional tooling for reading and browsing that result, not further evidence for or against the hypothesis.

Alongside the pipeline, a small web tool, the **Koselleck Machine**, makes the per-period networks directly explorable, without reading raw numbers. It has four views of the same underlying data.

Timeline. The primary view. Type a word and see, in one continuous strip, which community it belonged to in every period it appears in. Three encodings carry that information:

1. the community's plain-English label, with its lane (below) as a secondary colour cue;
2. an explicit gap where the word is simply absent from that period's vocabulary;
3. a marked step wherever the word's community actually changed from the period before, versus stayed the same.

Earlier drafts of this tool led with the graph explorer below, but historian collaborators who reviewed it found a force-directed diagram hard to read at a glance; what they consistently asked for instead was exactly this, which domain a word belonged to, over time, read directly rather than inferred from a picture. Clicking any period in the strip opens that word's full neighbourhood there, reusing the word search view below.

Ask. A conversational front end to the same measured results, for a question in plain language instead of a chart: type something like "did word meaning change the most around 1770-1830?" and get an answer grounded in the same networks, communities, and metrics described above, never freehand generation. A model answers only by calling a fixed set of tools that read the built data - it never sees raw tables, only what those tools return - so a question the data cannot answer gets an honest "the structure doesn't show that" instead of a guess, and every claim is tagged by how reliable that evidence actually is.

Word search. The same underlying data, presented as a plain table instead of a diagram: type a word and a period, and get back its closest neighbours by similarity, each neighbour's community, and whether the searched word's own community changed since the previous period. Useful when the actual numbers matter more than the picture, and reused directly by the timeline's own per-period drill-in above.

Graph explorer. A period slider drives a force-directed network diagram: nodes are words, their size reflects how connected they are, and their color marks which community they belong to in that period. A search box focuses the view on one word and its neighbourhood; moving the slider one period at a time shows that neighbourhood evolving, and a change in a word's color between periods marks a change of community membership. A side panel explains the method in plain terms and shows, for the currently selected resolution, how the migration measurement holds up across all fifteen resolution settings - the same sweep described above. Kept for browsing the raw network structure, and because building it first is what surfaced this whole labeling and lane apparatus, but no longer the primary way to read the tool, per the reception above.

Region toggle. Where the per-region runs described above have been built, both views also offer a region switch (combined, British only, American only) so the same word in the same period can be compared across the sub-corpora directly, and a shift can be checked against the possibility that it only exists in the mixture. The switch lists only the regions that were actually built, read off the data itself rather than assumed.

Interface scope, going forward. Of the four views above, Timeline and Ask carry the most weight in practice: Timeline for tracking one word's story across periods, Ask for querying the measured findings directly in plain language. The graph explorer gives comparatively little insight by comparison - a force-directed layout of hundreds or thousands of nodes doesn't tell a viewer much more than the same neighbourhood already shown as a table, and it becomes less useful, not more, exactly as the network grows large enough that the full graph can no longer be usefully viewed at once. Simplifying the interface down to Timeline and Ask, dropping or de-emphasizing the graph explorer, is under consideration.

Every community shown in any view carries a short, plain-English label (for example "Government & Law", "Moral Character & Virtue"), read by a language model from each community's most representative words: a convenience for a human reader, not a finding in itself.

A label is not re-read independently every period. Whenever the same correspondence step the migration fraction is built on maps a community back to a predecessor with a label, that label is a candidate to carry forward - but only after two independent model reads, given the inherited label and the community's current words, both agree it still fits; either reader saying no forces a fresh read instead. Across the three current snapshots ("Where the published labels live" below), 58–63% of labels in the combined and British runs carry forward this way; in the thinner American run, where more periods are a lineage's own first appearance, fresh reads are the majority instead.

Two-fifths to a third of communities in every region are sorted into the **"Structural / Uncertain"** lane rather than a topical one: the corpus is multilingual (English, Latin, Law French, Welsh, Scots, and more) and early modern, and a community detected by this method sometimes groups words by shared grammar (verb conjugations, comparative forms, spelling variants) rather than by shared topic. Every view surfaces this lane directly next to the label.

Each community is additionally sorted into one **lane**: a small, fixed list of fifteen broad domains (government and law, religion, medicine, trade, and so on), plus one explicit "structural / uncertain" lane for exactly the "mixed" case above. The lane list itself was authored once by hand, from a read-through of every non-mixed label already produced, then fixed. A language model assigns each community to one of these existing lanes; it never invents a new one per run. That fixed-in-advance property is what lets a lane stay a meaningful, comparable signal across independent pipeline runs even though, as the last section explains, the underlying community structure and the raw label text do not reproduce identically run to run. In the tools above, a community's plain-English label and number stay the primary signal; its lane is a secondary colour cue layered on top, not a replacement for reading the actual label.

The paragraphs above describe what the labels are for. The following four subsections give the mechanics, so that a technical reader can see exactly what the model was asked to do and reproduce or replace it.

3.1 Choosing the labeling resolution

Community detection produces a partition at every resolution in the sweep; the webapp displays and labels only one of them, a choice separate from the sweep itself. Today that display resolution is picked automatically: for every period and region-split variant independently, a bisection search - starting from resolution 1.0 and doubling or halving to bracket, then narrowing - finds the lowest resolution at which no community grows past a fixed 1,500-word cap. This search is independent of the fifteen-setting sweep described in item 5 above, not restricted to picking one of those fifteen values, so a given period or region can land anywhere, including above 16.0 or below 0.1. `docs/pipeline_manual.pdf`'s Stage 6 section covers the search mechanism and the one edge case it took a full production run to surface.

That automatic rule replaced an earlier, manually-tuned fixed value, used the same way for every period and region. The display resolution was originally set to 1.0 project-wide. Adding the British Library supplement later grew some periods' vocabularies past 170,000 words, and at 1.0 the largest community in the latest periods had grown with them to 24,608 words (14% of that period's vocabulary) - a size at which no label, model or human, could describe what was actually inside it. This was not a new failure mode introduced by the supplement, only a more visible one: the largest community had already been a stable 12-15% of the vocabulary at every period size tested, small and unremarkable at 70,000 words, absurd at 170,000.

Modularity falls off smoothly as resolution rises, with no sharp elbow in the tested range, so there was no purely statistical answer for the right resolution - the real ceiling was practical, how many communities per period stay browsable, not mathematical. The fixed value was raised by a concrete before/after check rather than a formula.

Take the community containing the word "system" in 1870-1890, the single worst offender at the time: at resolution 1.0 its top words read as an unstructured mix (*perjury, arrogance, indigestion, sensuality, bronchitis, extortions, dyspepsia*); at 2.0 and 3.0 the same words were still there, effectively unchanged; only at **4.0** did the community actually split, and the half that kept "system" became a real, nameable theme (*arrable, assessments, registration, payment, loans, forfeitures, coroners, judiciary, treasurers, sheriffs, freehold, manors, copyhold*) - civil administration and property law.

The same check on two other periods at 4.0 (one small, one mid-sized) produced equally coherent results (philosophy/rhetoric/mathematics; church governance and clergy), so the fixed value was raised project-wide to 4.0 rather than only for the largest periods, and `leiden.resolution_sweep` was extended to match at the time (up to 4.0, from 2.0) so the manually-chosen display value and the tested range stayed aligned.

That manual approach was itself a stopgap: growing the corpus further would only have called for raising the fixed value again and again, since the right resolution scales with vocabulary size rather than being a constant. It was later replaced entirely by the automatic, per-variant rule described above, and `leiden.resolution_sweep` was extended again to its current range, fifteen settings from 0.1 to 16.0 (item 5), as part of that same move - so today's display choice and the tested range stay aligned by construction rather than by manual bookkeeping.

3.2 The prompt

The prompt lives in the repository as a single Markdown file, currently `src/prompts/community_labeling_v3.md`, which is parsed at runtime by `src/label_communities.py` rather than duplicated as a string constant in code, so the file stays the one place the instructions and the lane list are edited. It has three sections: the fixed lane list, the system prompt, and the user message template. One model call is made per Leiden community, at the auto-picked display resolution described above.

The system prompt, quoted from that file, tells the model:

> "You are labeling word clusters from a computational study of an early modern through nineteenth-century English-language corpus: the Text Creation Partnership (EEBO, ECCO, Evans; 1500-1800, clean manually-keyed transcription) plus the British Library's digitised 19th-century books collection (1800-1900, OCR text). Each cluster is a 'community' found by running the Leiden algorithm on a per-period word-similarity network, words that are used in similar contexts more than they are used with the rest of that period's vocabulary. The corpus is multilingual (English, Latin, Law French, Welsh, Scots, and other languages appear untranslated) and pre-modern, so a community sometimes groups words by shared grammatical form (verb conjugations, comparatives, spelling variants), by shared foreign language,

> or by proper names (people, places, biblical figures) rather than by a real topic. In periods drawn from the British Library supplement, a community can also be a cluster of OCR fragments - words with a missing prefix or suffix from a misread hyphen or scan artifact - rather than real words at all. *All of these are honest and expected outcomes when they are genuinely what the community is, not a failure to avoid.*"
>
> It then asks the model to work through, in order, before deciding anything:
>
> 1. what the community's most central (*Core*) words suggest on their own;
> 2. whether its less-central (*Mid-rank* and *Peripheral*) words support or undercut that reading, naming at least one specific word from each tier; and
> 3. only then, whether a genuine topic holds across all three, or whether the community is better described as grammatical, foreign-language, or name-driven.
>
> That ordering is itself the point of a required `reasoning` field, a later addition to the prompt made for exactly this reason (see below). Once that reasoning is done, it asks for exactly three decisions:
>
> 1. "Whether most of the words shown - across every tier given, not just the Core - share a genuine topical or thematic connection a historian would recognize (e.g. law, medicine, religion, trade), as opposed to being grouped mainly by grammar, a shared non-English language, or being mostly proper names or fragments."
> 2. "A short plain-English label, two to five words, for what the community is actually about. If the grouping is grammatical, foreign-language, or name-driven rather than topical, the label should say what kind of cluster it is (e.g. 'Second-Person Verb Forms', 'Latin Text', 'Biblical Names'), not force a topic onto it."
> 3. "Exactly one lane from the fixed list above. Use 'Structural / Uncertain' whenever the answer to (1) is no, do not stretch a grammatical, foreign-language, or name cluster into a thematic lane just because a few of its words are suggestive of one."

The required `reasoning` field is a direct fix for a specific failure mode, not a general polish pass. An earlier version of this prompt asked for the same three decisions with no forced reasoning step, and too often reached for an easy grammatical or "Structural / Uncertain" read from the Core tier alone, without evidence the Mid-rank and Peripheral words had been weighed at all. Ordering `reasoning` before `label` and `lane` in the tool schema the model must call forces that weighing to happen.

A blind check on two real communities, each sent to a fresh, context-free model twice, once with each version of the prompt, found the earlier version defaulting to "Structural / Uncertain" both times despite its own draft label already naming a real theme, and the current version reaching a concrete, evidence-grounded topical lane both times instead.

The user message carries only the material the model is meant to judge, filled in from the extraction step: the region, the period, the community's size in words, and a sample of its member words by network-degree rank. That sample used to be a flat list of the 25 words with the highest degree in the community, its most central members, until a real gap surfaced: a flat top-slice only ever shows words tightly bound to a community's core, hiding a community whose core looks coherent but whose lower-degree tail has drifted to a different topic entirely.

The sample is now stratified instead: up to 15 words each from the top (Core), middle (Mid-rank), and bottom (Peripheral) of the ranked membership list, 45 words in total for any community large enough to split three ways; a smaller community is shown in full. The `reasoning` field described above exists specifically so a model has to look at all three tiers, not just the Core one this older flat sample used to show alone.

Two properties of this setup matter more than the wording. First, the lane vocabulary is closed. The model is not asked to invent a taxonomy; it must choose one of fifteen lanes fixed in advance, listed in the same file and reproduced here in the order the prompt gives them:

1. Government, Law & Administration
2. Religion, Theology & the Church
3. Morality: Virtue & Vice
4. Medicine, Body & Health
5. Science, Mathematics & Natural Philosophy
6. Nature, Landscape & Weather
7. Military & Warfare
8. Trade, Finance & Commerce
9. History, Genealogy, Nobility & Kinship
10. Rhetoric & Persuasion
11. Literature, Drama & Poetic Diction
12. Domestic Life, Dress & Household
13. Geography & Territory
14. Books, Learning & Scholarship
15. Structural / Uncertain

That list was not invented for the prompt either: it was derived by hand from 707 labels already produced across the combined, British, and American runs, then frozen. Second, the response is not free text. The model must answer by calling a tool, `assign_label`, whose schema takes a `label` string ("two to five word plain-English label") and a `lane` constrained by a JSON-schema `enum` to exactly the fifteen lanes parsed out of the prompt file. A returned lane outside that list is rejected and the call retried, so a label with an out-of-taxonomy lane can never reach the data files.

3.3 Generating and checking labels

`extract_community_words.py` writes, for each populated period at the configured display resolution, every community's size and its stratified Core/Mid-rank/Peripheral word sample (see above) to `communities/community_words_display[_<region>].json`, once for the combined corpus and once per region, since a region's own Leiden run has its own community numbering. `label_communities.py generate` turns that JSON into a human-editable CSV, walking periods in chronological order; `docs/pipeline_manual.pdf`'s Stage 7 section gives the full command sequence and CSV schema, and README's Labeling communities section has the same commands for a bare clone.

A community whose best correspondence to the period before it (the same correspondence step the migration fraction is built on) has a label is a *candidate* to inherit it, not an automatic inheritance: two independent model reads, given that inherited label and the community's current words, are asked the same yes-or-no question, does this label still fit. Only unanimous agreement carries the label forward unchanged; either reader saying no forces a fresh read, exactly as a genesis community (no predecessor at all) gets one.

This replaced an earlier, blunter safeguard, a fixed cap on how many periods a label could inherit before a forced re-read, itself built after a real case: a community genuinely holding Dutch and German text at 1510-1530 was still labeled "Dutch and German Text" fifteen periods later, by which point its content had passed through seven unrelated themes - including, at the end, English church governance - all unread. Checking the actual words on every candidate inheritance, rather than counting periods, catches that drift directly instead of bounding how long it can go unnoticed.

`compile` turns the CSV, including any hand edits, into `communities/community_labels_display[_<region>].json`, which is the exact file `webapp/app.py` reads. It also stamps a `_meta` block into that JSON recording the resolution, the region, the number of communities, how many were human-edited, how many were inherited, and a standing caveat that the labels are a single model read-through and not an empirically validated taxonomy. Compiling used to append a literal "(mixed)" to the label text of every `Structural / Uncertain` community; that is no longer done. The `lane` field already carries the “this is not a real topic” signal on its own, and stacking the word “mixed” onto an already-decisive reading (“Second-Person Verb Forms”, “Biblical Book Abbreviations”) made even a clear description look like the model could not settle on one.

`publish` copies the CSV and the compiled JSON for a region out of the data directory and into this repository. `--region` also accepts a region name or `all` at every subcommand.

3.4 Where the published labels live

The label files that the pipeline writes live in the data directory, which is gitignored and does not travel with the repository. That is why a collaborator cloning the repo previously saw no community names at all. The `publish` subcommand exists to fix exactly that: it copies a snapshot into the repository’s `labels/` directory, which currently holds one CSV and one compiled JSON per region. The filenames no longer embed a resolution number: since the display resolution is now picked per variant, not once globally (“Choosing the labeling resolution” above), a single number in the filename would be meaningless once different periods in the same run can land on different resolutions.

```
labels/community_labels_display.csv
labels/community_labels_display.json
labels/community_labels_display_british.csv
labels/community_labels_display_british.json
labels/community_labels_display_american.csv
labels/community_labels_display_american.json
```

That is the citable snapshot: the version of the labels this document describes, readable without running the pipeline or holding the corpus. `publish` only copies files to disk; it never commits, so the snapshot is reviewed by hand before it is added to version control.

4 Current limitations

Corpus coverage. 1800-1900 is now covered by the British Library supplement described above, but coverage is uneven in four ways:

1. ECCO is thin after 1780, so 1770-1790 leans on more American than British documents (Evans-TCP, 1,030, against mostly-ECCO’s 792).
2. The British Library supplement brings the next window, 1790-1810, close to parity instead.
3. The `american` region has no data at all past 1800, since Evans-TCP stops there and

the supplement is British-only (see Corpus above).

4. Per-region results carry wider uncertainty than the combined headline numbers, and the `american` region specifically cannot speak to anything past the Sattelzeit's closing edge.

OCR artifacts. The British Library text is OCR-derived, unlike TCP's manually corrected transcription. `ocr_refinement.py` repairs two classes of damage:

1. an original line-break hyphen left sitting in the text (*"utrum- que"* for *utrumque*);
2. the harder case of the OCR dropping that hyphen entirely and leaving a bare space instead (*"par ticulars"* for *particulars*), fixed by merging two adjacent words only when the merged form is real and at least one half isn't independently real on its own - checked by hand against the one period a reference wordlist covers, 1790-1810 (142 merges made in the first 120 changed documents, all confirmed correct).

That wordlist is built only from that period's own TCP text, so periods entirely after 1800 remain unrepaired and untested by this fix. A different, unrelated source of noise - French and Welsh text surfacing as its own network community - is fixed separately, by dropping a document once a language classifier is at least 90% confident it isn't English.

Word-level noise in the network. Two further patterns pass through into the trained networks and cluster tightly enough by co-occurrence to form their own communities, neither yet fixed at the stage that would need to change (which word becomes a node):

1. Latin citation-apparatus abbreviations (*capvt*, *deute*, *regu*), correctly transcribed originals, not damage of any kind, but carrying no topical content either.
2. A distinct, unconfirmed TCP transcription pattern, surfaced separately when the labeling resolution was raised, in a handful of communities from 1510-1650: a missing "con-"/"com-" prefix on otherwise-recognizable words (*mytted* for *committed*, *uersation* for *conversation*, *monwealth* for *commonwealth*), most likely a scribal abbreviation mark mishandled during text extraction - not confirmed against the raw XML, and no fix attempted.

`docs/pipeline_manual.pdf`'s Stage 5 section has more detail on the first of these.

Label reliability. A label is a candidate to inherit from its predecessor only after two independent model reads agree it still fits the community's current words ("Generating and checking labels" above); either reader saying no forces a fresh read. Across the current label snapshots, that leaves 37–42% of labels in the combined and British runs, and the majority in American, as an independent read for a given pipeline run. That read is still a single, unvalidated pass by a language model, not a checked ground truth - useful for orientation, not for citation.

Reproducibility. Rerunning the full pipeline from scratch on the same corpus does not reproduce an identical result. Community detection itself is properly seeded and deterministic given a fixed network; the gap sits one step earlier, in training. Word2Vec training here uses multiple parallel threads for speed, and that kind of parallel training is not bit-for-bit repeatable even with a fixed random seed, because the exact order in which threads update shared values depends on how the operating system happens to schedule them that run. Small differences in the trained word vectors cascade into a similarity network with slightly different edges, which cascades into a structurally similar but not identical community partition - in one earlier comparison, two full reruns of the same combined corpus produced community counts a few apart.

This does not bear on the claim under test: reorganization is measured by whether `migration_fraction` concentrates at the Sattelzeit, checked across all fifteen resolutions within a single run, never by whether a specific community keeps its id or exact member words across two independently retrained copies of the pipeline. The specific migration-fraction figures reported in the Result above (0.41 through 0.62) are exactly this kind of point estimate: expect them to drift somewhat, not to reproduce bit-for-bit, on an independent rerun. The one thing meant to stay comparable across independent reruns is the fixed fifteen-lane list, precisely because it is authored once by hand and never reinvented per run.

5 Where this leaves the hypothesis

Koselleck's claim was that the Sattelzeit was a moment when concepts reorganized *as a system*, not just individual words drifting independently. The network test built here finds a real, gradual rise in migration fraction across the transitions that open and close the 1770-1830 window - a direction consistent with that claim - but not a spike isolated to the window: the rise continues past 1830 rather than falling back, and a larger, likely-artifactual rise appears earlier, at 1690-1730 (item 7). That combination is what "does not show a clean, isolated reorganization spike" means in practice: the evidence leans toward Koselleck being right about direction, without being strong enough on its own to call the case closed. The result is suggestive, not confirming, and rests on this network test alone.